



MAVEN MARKET DATA ANALYSIS: FINAL PROJECT

Unlocking Retail Insights Through Data



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Domain :

Maven Market is a retail sales dataset containing transactions, customers, products, stores, regions, calendar dates, and product returns .

It can be used to analyze sales performance, customer behavior, product trends, store/region performance, and returns.

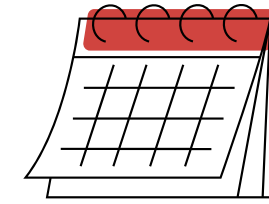
Project Objective :

- Analyze overall sales and revenue performance over time.**
- Identify top-performing products, stores, and regions.**
- Analyze customer purchasing behavior and trends.**
- Evaluate product returns and identify return patterns.**
- Compare sales performance across different regions and stores.**
- Identify key trends and insights to support business decision-making.**



Dataset Overview

[dataset link](#)



from 1997 to 1998



~250K

trans



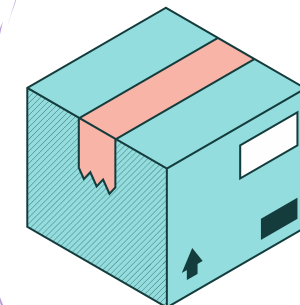
24

Stores



~10K

Customers



1560

Products

Major Analytics business area

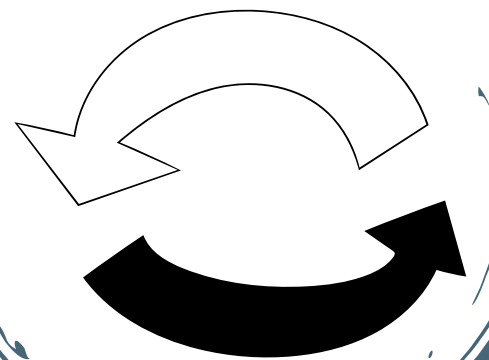
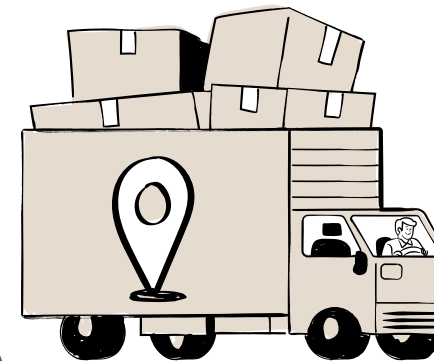
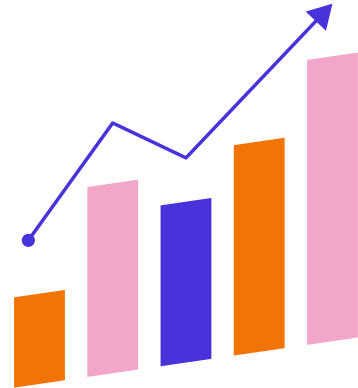
Customers

Sales

Trend

Products

**Return &
Service**



Business Flow :

**“It all starts with a customer need:
Can we be the brand that comes
to their mind? From discovery and
purchase to delivery, satisfaction,
and even return — every step tells
us something about our business.”**



01

Overall Business Performance

- Revenue, Cost & Profit
- Profit Margin %
- Transactions & Units Sold
- Revenue & Profit Growth
- Average Order Value

02

Product Performance

- Top & Bottom Products
- Revenue & Profit by Product/Brand
- Units Sold & Profit Margin
- Product Return Rate

03

Customer Analytics

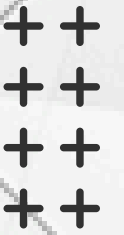
- Customer Spending & Purchase Frequency
- New vs. Repeat Customers
- Customer Retention
- Revenue per Customer
- Customer Segmentation

04

Returns Analytics

- Return Rate & Returned Units
- Returns by Product, Brand & Store
- Return Trends
- Revenue Impact of Returns

Business Problem



➤ Sales Performance

Inconsistent revenue across regions
Unclear top-performing products

➤ Customer Trends

Low customer retention rates
Limited segmentation insights

➤ Inventory Optimization

Excess stock in low-demand categories
Supply chain inefficiencies

➤ Objectives

Identify growth opportunities
Improve data-driven decision-making





Data Cleaning & Preparation

Initial Data Integration (Query Appending)

- Merged MavenMarket_Transactions_1997 and MavenMarket_Transactions_1998 into a consolidated facts table: MavenMarket_Transactions_All.

Handling Missing Values

1

- 1.1. Identified missing values in the recyclable and low_fat columns.
- 1.2. Replaced all missing entries with NULL to maintain data integrity.
- 1.3. Identified and addressed a single missing value (NULL) in the last_name column within MavenMarket_Customers.

2

Removing Duplicates

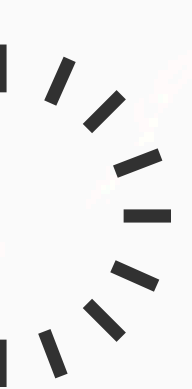
- 2.1. MavenMarket>Returns_1997-1998: Identified and removed 5 duplicate records.
- 2.2. MavenMarket_Transactions_All: Identified and removed 9 duplicate records.

3

Correcting Data Types

- 3.1. MavenMarket_Stores: Converted last_remodel_date from INT to DATE format.





Data Cleaning & Preparation

4

Outlier Detection

- 4.1. Performed statistical checks on numeric distributions across all facts and dimensions.
- 4.2. Found 0 outliers requiring treatment or capping.

5

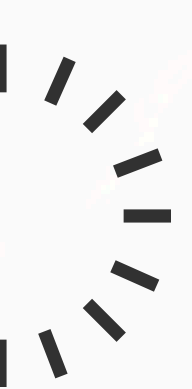
Validating Logical Integrity (Invalid / Impossible Values)

- 5.1. Verified that Quantity, retail_price, and cost values are strictly greater than or equal to 0.
- 5.2. MavenMarket_Transactions_All: Validated transaction dates fall within 1997–1998 and ensured stock_date is before or equal to transaction_date.
- 5.3. MavenMarket_Stores: Ensured last_remodel_date is after or equal to first_opened_date.
- 5.4. MavenMarket_Stores: Verified that grocery_sqft is strictly less than total_sqft.

6

Standardizing Formats & Categories

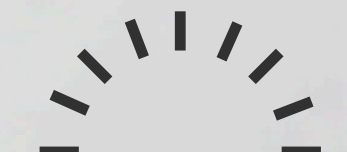
- 6.1. Reviewed all categorical columns; no standardization or formatting changes were required.
- 



Data Cleaning & Preparation

Feature Engineering & Column Transformations

7

- **7.1. MavenMarket_Calendar:** Extracted Year, Quarter, Month, and Day attributes for complete Time Intelligence analysis.
 - **7.2. MavenMarket_Transactions_All:** Created Stock_Lead_Time_Days to calculate the difference between stock_date and transaction_date.
 - **7.3. MavenMarket_Stores:** Created Days Since Opening to Remodel to calculate the difference between first_opened_date and last_remodel_date.
 - **7.4. MavenMarket_Customers:** Transformed the categorical yearly_income text ranges (e.g., "\$30K - \$50K") into a numeric Average_Income column to enable accurate numerical aggregations and correlation analysis.
 - **7.5. MavenMarket_Stores:** Calculated Non-Grocery Sqft by subtracting grocery_sqft from total_sqft.
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SQL Analysis Findings

01 Overall Business Performance

Sales, Profit & Basket Size

- Total Sales: **\$1.76M** total **revenue** with **\$1.05M** profit.
- Healthy Profit Margin: High and **steady at 59.67%**.
- Small Basket: Customers **spend an average of \$6.54** per visit.

Customers visit often to buy cheap daily items, but we earn a strong 60 cents profit on every dollar.

Revenue Over Time

- 1997 Baseline: Monthly sales stayed flat **between \$42K and \$57K**.
- Big 1998 Jump: Sales jumped to **\$98K in Jan 1998** and stayed high all year.
- All-Time Peak: Closed at **\$120,161** in the last quarter (December) 1998.

Sales did not just rise for holidays; the business doubled in size starting in 1998 and kept that level.

Seasonality & Growth

- Yearly Growth: Revenue jumped by **+112% from 1997 to 1998**.
- All Top 5 Months: Every single top month happened in 1998.
- All Bottom 5 Months: Every single low month happened in 1997.

Growth is strong and real across all 12 months, not just a one-time lucky spike.

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Product Performance

Revenue & Profit Leaders

- Dual Leader: Hermanos Green Pepper leads both revenue (\$2,490) and profit (\$1,671).
- Top Products: Hilltop Mint Mouthwash (\$2.45K) and Head Cheese (\$2.42K).

The business does not rely on one big product; sales are safely spread out across the whole store.

Margin Watch & Brands

- Best Brand: Hermanos is the #1 brand in sales (\$56.7K).
- Low Profit Items: Some popular items (like Hot Dogs and Cheese) make less profit (~51%).
- Issue: They make about 8% less profit than the store average (60%)

Returns & Dead Stock

- Low Returns: Very few items are returned (all under 3%).
- Dead Stock: CDR Grape Jelly made \$0 with zero sales.
- Slow Items: Other CDR products sell poorly (under 400 units).

Returns are fine, but we should remove useless CDR items to clear shelf space.

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03

Store & Regional Trends

Store Revenue & Profit

- Top Stores: **Store 13 (\$170.4K)** and **Store 17 (\$157.7K)** make the most sales by far.
- Top Profit: The same two stores also generate the highest profit.
- Healthy Margins: Profit margin remains strong at **~59.6%**.

Regional Disparities

- Underperforming Region: Central West sales are extremely low at just **\$9.3K**. While Other regions perform ranging **from \$61K to \$108K**.
- Low Returns Everywhere: Returns stay tiny across all areas (**~1.1%**).

Store-Level Variances

- Big Local Gap: In the same region, Store 13 is far above **average (+\$76K)**, while Store 2 is far **behind (-\$85K)**.
- Store Issue: Low sales come from bad store location or management, not low customer demand.
- Steady Margins: Leading stores in all areas keep the same **healthy profit (~59.6%)**.

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- 1. Who are Maven Market's most valuable customers?**
- 2. Which customer demographics generate the highest revenue?**
- 3. What is the average customer spend (monetary)?**
- 4. How frequently do customers make purchases?**
- 5. What percentage of customers are repeat customers?**
- 6. Which customer segments contribute the most to revenue and profit?**

KPIs :

- Total Customers • Active Customers • Repeat Customer Rate • Average Customer Spend**
- Average Purchase Frequency • Top Customer Spend • Top Customer Profit**

1. What percentage of purchased products are returned?
2. Which products have the highest return rates?
3. Which stores have the highest return rates?
4. Are return rates increasing over time?
5. Which brands or product categories contribute most to returned units?
6. How much revenue is potentially affected by returns?

KPIs :

- Total Returned Units • Return Rate % • Returns by Product • Returns by Store • Returns by Brand
- Return Trend • Revenue Impact of Returns

Maven Market — Power BI Dashboards

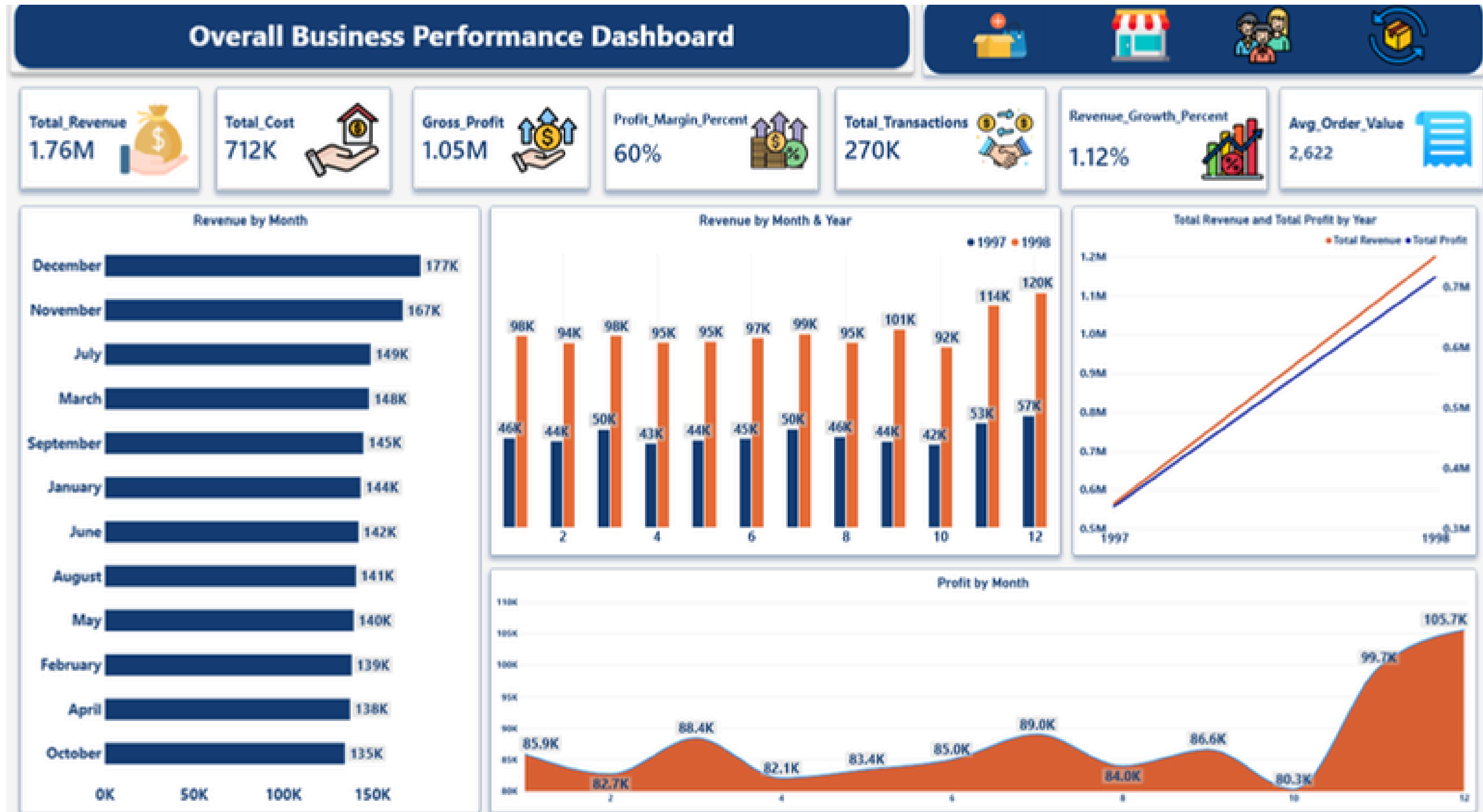
5 dashboards

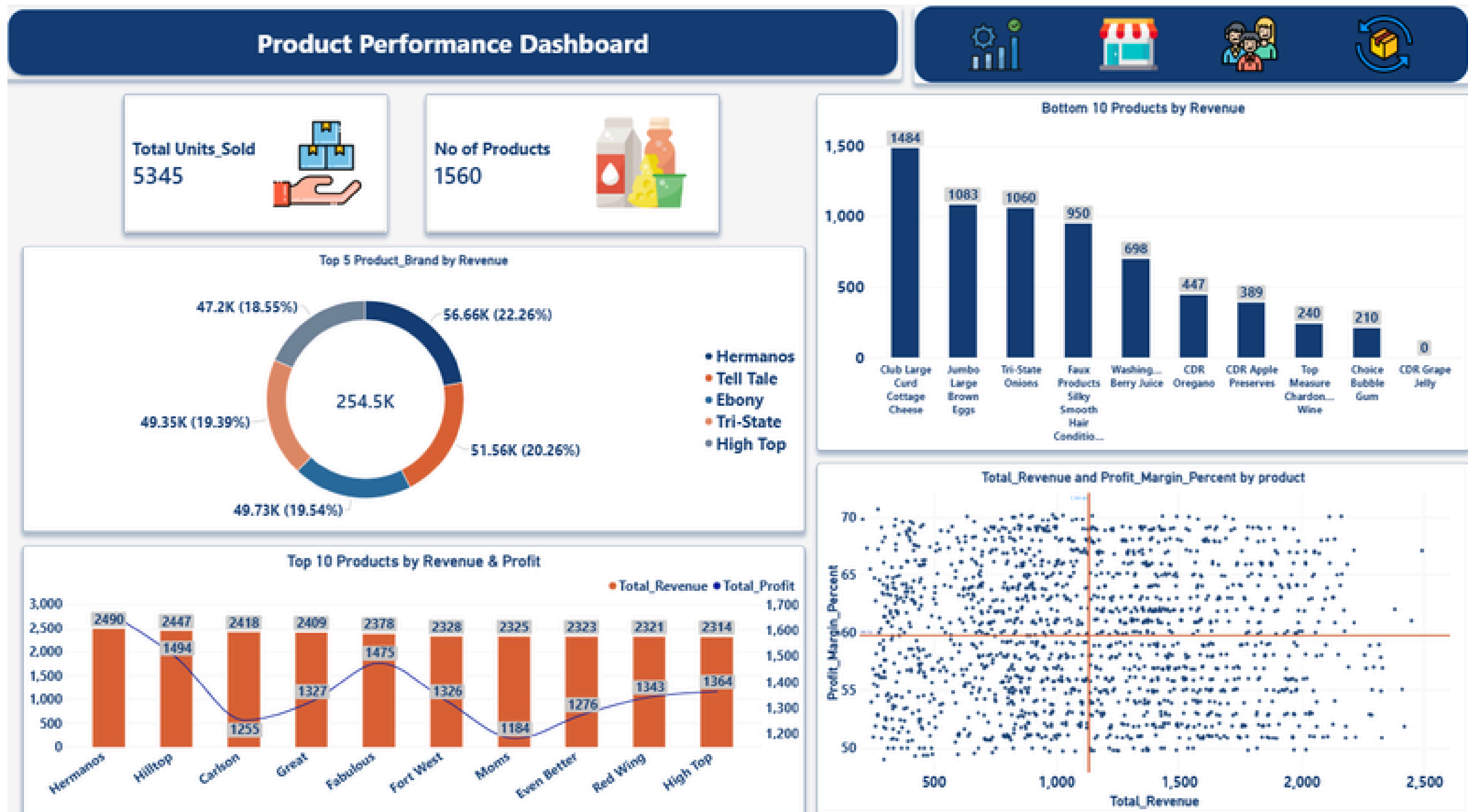
- 1 Overall Business Performance
- 2 Product Performance
- 3 Store & Regional Performance
- 4 Customer Analysis
- 5 Product Returns & Revenue Risk Dashboard



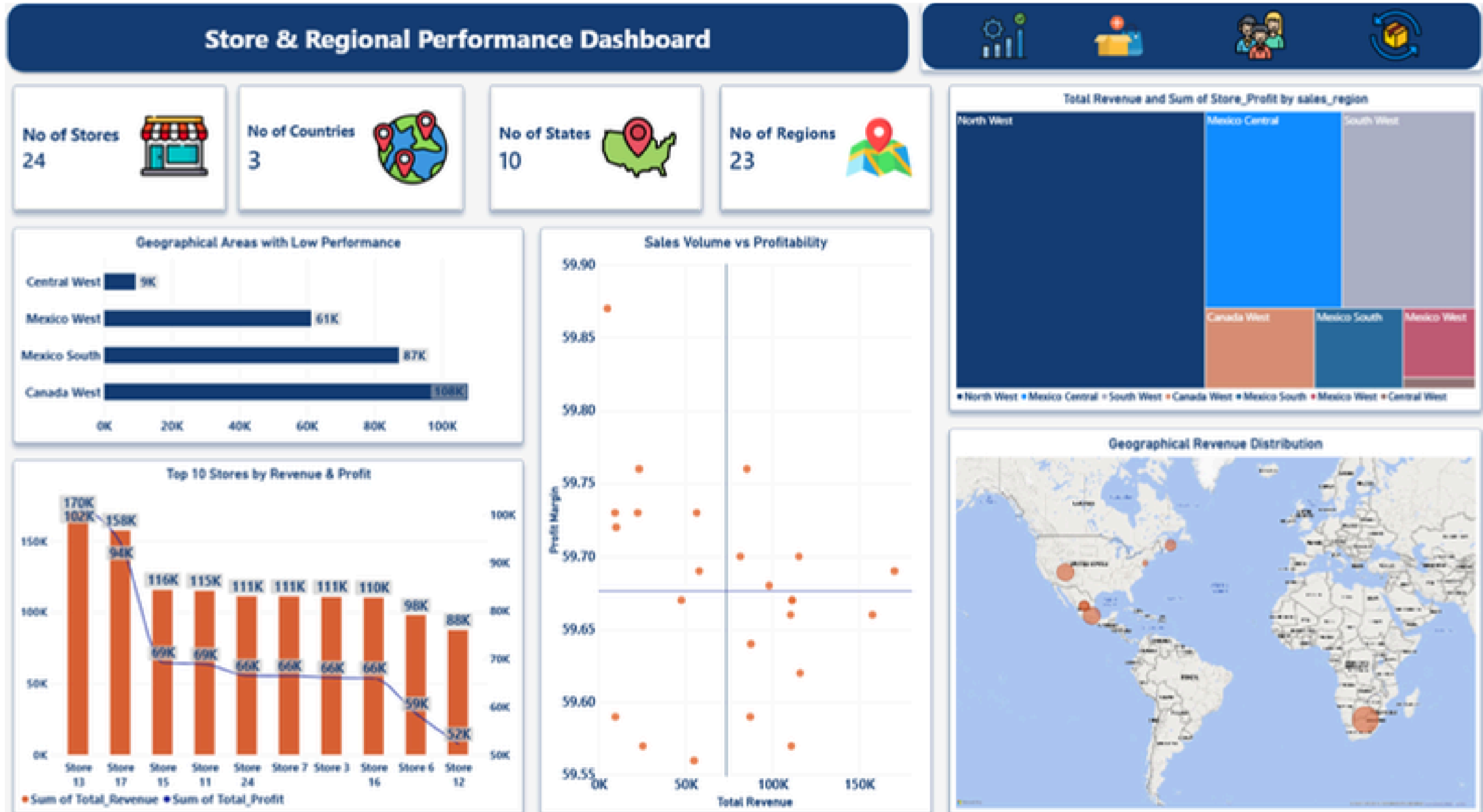
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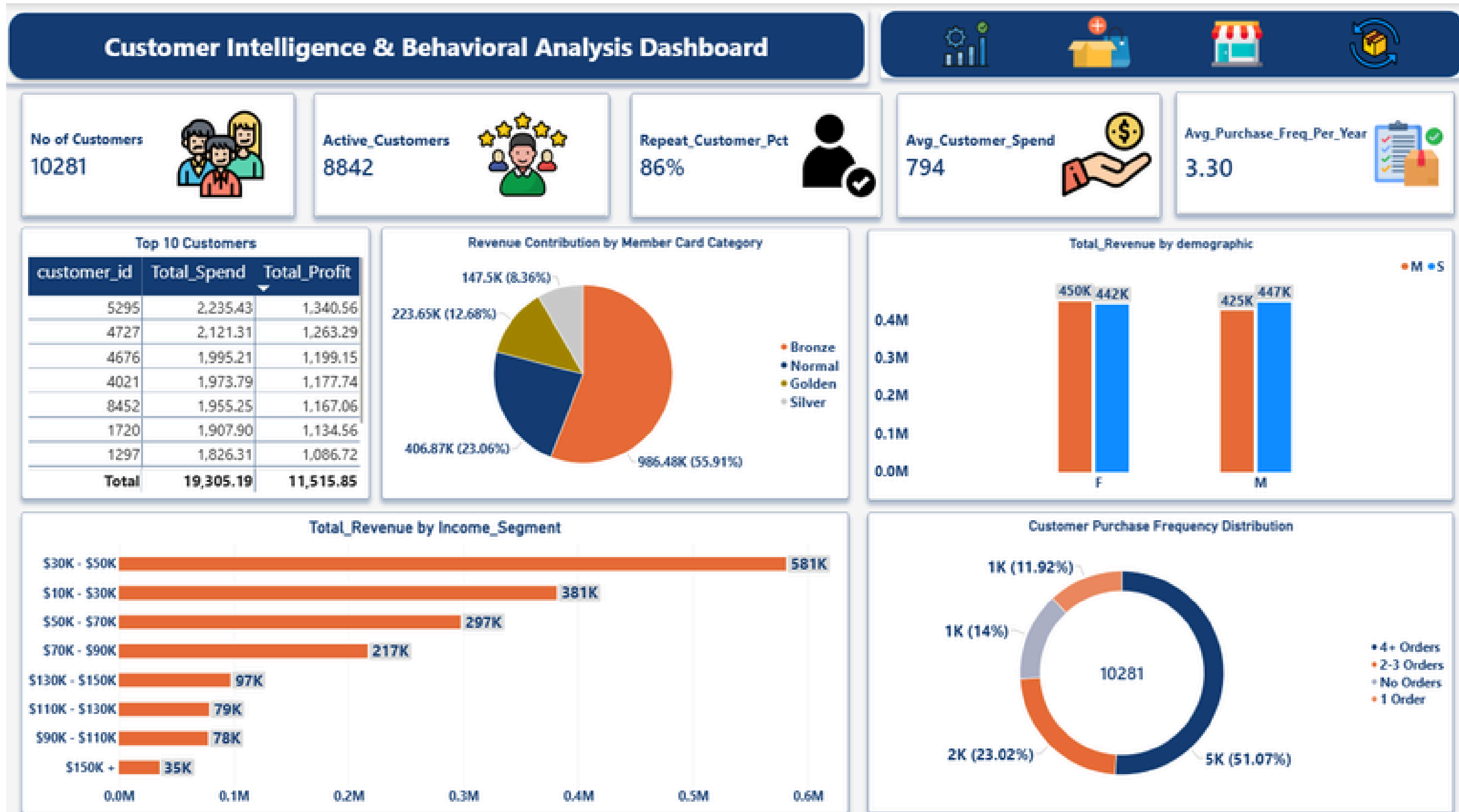
Overall Business Performance Dashboard



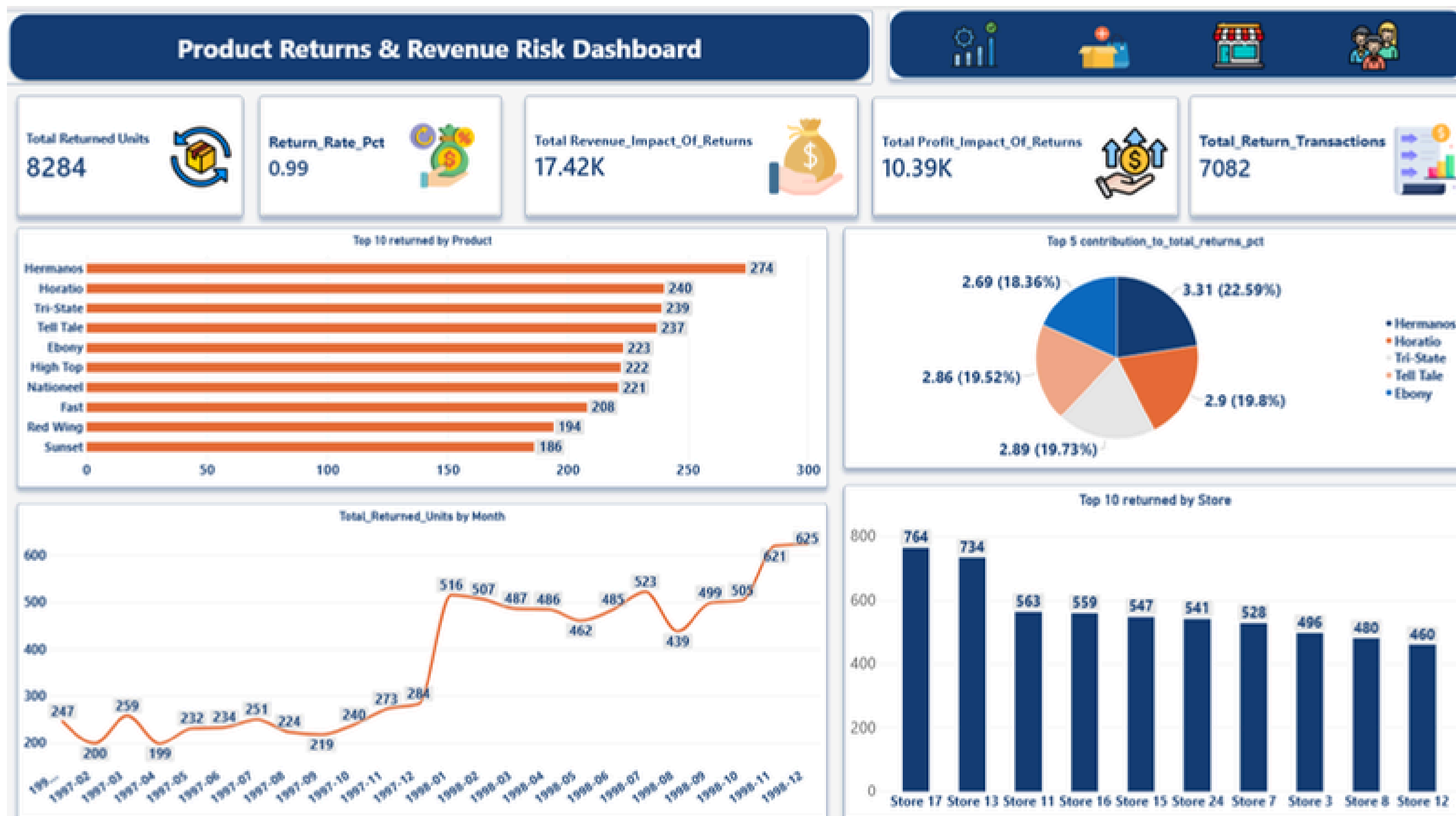


Store & Regional Performance Dashboard





Product Returns & Revenue Risk Dashboard



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Key Business Insights

1998 revenue and profit were both up more than 1.12% vs. 1997.

Revenue growth 1.12%
and profit growth 1.12%



Seasonal sales peaks in Q4

December is the peak month: \$177K total revenue, with \$120K in Dec 1998



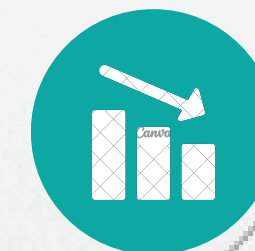
Regional performance gaps

Central West is the weakest area at \$9K; North West leads



Customer loyalty drives 60% of revenue

86% repeat buyers; 51.07% place 4+ orders



Top 10 products = 40% of total sales

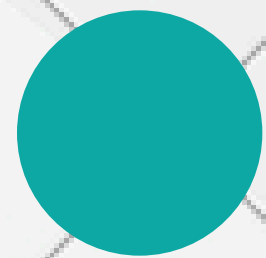
Many high-revenue products sit below the ~60% mean margin

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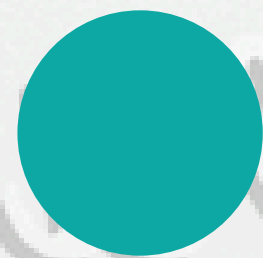
Where the Opportunity Is

The strongest signals from products, stores, customers and returns



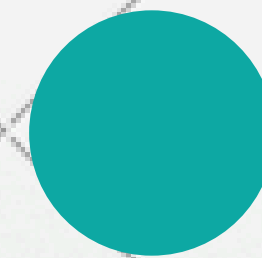
PRODUCTS

- 1,560 products sold across the period
- Hermanos leads revenue; Fabulous outperforms on profit
- Weak products include CDR Grape Jelly and Choice Bubble Gum



STORES & AREAS

- Store 13 and Store 17 lead revenue and profit
- North West is the strongest region
- Central West is only \$9K in revenue



CUSTOMERS & RETURNS

- \$30K–\$50K income group generates \$581K
- Bronze members drive 55.91% of card-linked revenue
- Returns concentrate in top-selling products and stores

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Data-Driven Recommendations

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01

Protect peak season

Plan stock, staffing and promotions

02

Fix high-sales / low-margin products

Review cost, price and pack size on products below mean margin

03

Lift weak areas

Audit weak areas for pricing, hours and staffing, then copy wins from leading regions.

04

Grow loyalty and basket value

Use cross-sell and early-access offers to increase order value

05

Cut and track return drag

Audit return reasons for Hermanos, Horatio and Tri-State; focus on Stores 17 and 13

Priority lens

Improve profit quality without sacrificing growth.



Thank You !

